# F79SP Session 6: Other Variants of Random Walks (1)

## 1 Other random walks

We demonstrate the usage of knowledge of SRW to solve problem involving other kind of random walks.

**Simple random walk with possibility of zero jumps**

Consider the random walk $\{S_n\}_{n\geq 0}$ on integer state space $\mathbb{Z}$. Let $S_0 = a > 0$ and $S_n = a + \sum_{i=1}^{n} \xi_i$ for i.i.d. $\xi_i$'s with: $\mathbf{P}(\xi_i = 1) = p$, $\mathbf{P}(\xi_i = -1) = q$ and $\mathbf{P}(\xi_i = 0) = 1 - (p + q)$, where $p + q < 1$.

- This random walk can be studied by considering a modified process obtained by discarding the zero jumps steps, giving a SRW $\{S'_n\}_{n\geq 0}$.

- For the SRW, let $\mathbf{P}(\xi_i = 1) = p'$ and $\mathbf{P}(\xi_i = -1) = q'$ with $p' + q' = 1$.
- $p'$ is thus the probability of jump up, given that a jump occurs, i.e., $p' = \frac{p}{p+q}$. Likewise, $q' = \frac{q}{p+q}$.

**Ruin problem**

We study the ruin problem of $\{S_n\}_{n\geq 0}$.

- Ruin occurs for $\{S_n\}_{n\geq 0}$ if and only if it occurs for the SRW $\{S'_n\}_{n\geq 0}$. Hence,

$$\mathbf{P}(R) = \min\left(1, \left(\frac{q'}{p'}\right)^a\right) = \min\left(1, \left(\frac{q}{p}\right)^a\right)$$

- If $p < q$, then we can show that $\mathbf{E}(T_R) = \frac{a}{q-p}$ (**Exercise!**).